

Alireza Amiri

MECHATRONICS ENGINEERING STUDENT

Tehran, Iran

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“Focused on tackling challenges with innovative solutions and boundless enthusiasm for learning.”

Objective — Driven to solve challenges with creativity and determination, combining a passion for learning with the ability to navigate ambiguity and deliver results.

Education

K. N. Toosi University of Technology University

Tehran, Iran

M. S. STUDENT IN MECHATRONICS ENGINEERING

Sep. 2023 – Present

- GPA: 18.61/20 ()
- Thesis: Design and Implementation of a Magnetically Levitated Planar Motor

Shahed University

Tehran, Iran

BACHELOR OF SCIENCE IN ELECTRICAL ENGINEERING (CONTROL ORIENTATION)

Jul. 2019 – Jul. 2023

- GPA: 16.03/20
- Thesis: Design and Implementation of a PID-based Control System for Achieving Stable Magnetic Levitation

Experience

Moneyar

Tehran, Iran

RESEARCH AND DEVELOPMENT – DATA SOLUTIONS AND BI

Feb 2024 – Present

- Led the research and development of data entry systems as foundational components for automated BI solutions.
- Designed and implemented processes for transforming raw data into actionable insights through advanced BI tools.
- Collaborated with cross-functional teams to explore innovative methodologies, ensuring cutting-edge data solutions.

Behnood Pharmed

Tehran, Iran

DESIGN AND DEVELOPMENT ASSISTANT

Aug 2023 – Oct 2024

- Gained practical insights into business processes by designing and developing management dashboards.
- Applied financial and project management concepts to real-world scenarios, enhancing professional growth.
- Utilized expertise in databases and QlikView to contribute to data-driven decision-making tools.

MAPNA Generator Engineering & Manufacturing Co.

Karaj, Iran

MAINTENANCE AND REPAIR ENGINEERING INTERN

Jun. 2022 – Sep. 2022

- Gained insights into industrial large-scale factories and basic performance of industrial machinery.
- Learned fault diagnosis procedures and cost-efficient problem-solving techniques.
- Experimented with industrial control systems, including PLC and LOGO systems.

Haftonim Publication

Remote

TRANSLATOR

Mar. 2021 – Jul. 2021

- Translated a scientific book, “Archimedes and the Door of Science,” for teenage readers.
- Developed skills in literary translation and adapted text to Persian.

Self-Employed

Remote

NFT MARKETING

Jan. 2022 – Jul. 2022

- Launched and marketed “Callipen,” a calligraphy artwork brand.
- Achieved international sales and developed cross-cultural communication skills.

Selected Projects

Waypoint Generation based on Crop Row Detection Using Unet and Hough Transform

A PROJECT OF MECHATRONICS COURSE

Jan. 2024 – Sep. 2024

- Develops a method to generate precise waypoints for autonomous agricultural robots using aerial imagery.
- Utilizes U-Net based image segmentation for crop detection and Hough Transform for row detection, followed by a mathematical conversion to global coordinates.
- The proposed system achieved high accuracy in crop and row detection, enabling precise waypoint generation for autonomous robot navigation.

Magnetic Levitation

B.Sc. THESIS

Mar. 2022 – Sep. 2023

- Developed an Arduino-based PID controller to achieve stable magnetic levitation.
- Experimented with PID and H-infinity control methods using MATLAB and Simulink.
- Designed and optimized an electrical circuit to support system stability.

Speech Emotion Detection

DIGITAL SIGNAL PROCESSING PROJECT

Jul. 2022 – Aug. 2022

- Trained a machine learning model to classify emotions and gender from Persian speech data.
- Achieved 60% accuracy in the best-case scenario.

Publications

Waypoint Generation based on Crop Row Detection Using Unet and Hough Transform Accepted paper in ICROm 2024

Skills

Programming Languages	Python,Arduino, Verilog, SQL, C++
Engineering Tools	MATLAB, Simulink, ROS2, Proteus, Ansys, ISE, Linux
Soft Skills	Creativity, Problem Solving, System Thinking, Adaptibility, Teamwork, Documentation

Languages

Persian	Native
English	Fluent
Arabic	Conversational